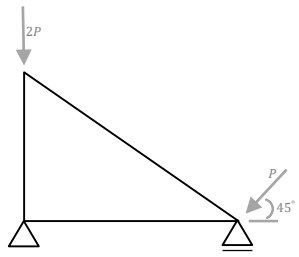
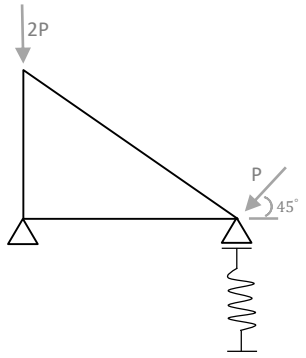
<u>تمرین سری پنجم</u>(روش های المان محدود)

(دانشجو: سبا آزادگان / شماره دانشجویی: ۶۰۲[redacted]۹۷)

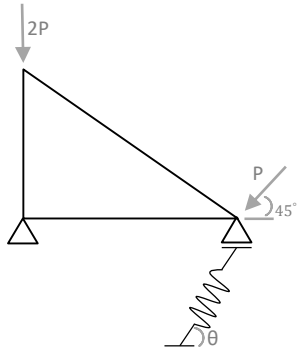
1. **بردار نیروها و تغییر مکان های دو المان دیگر سازه را محاسبه کنید. آیا تعادل برقرار است؟**
2. **سازه داده شده را برای بارگذاری مفروض آنالیز کنید.**



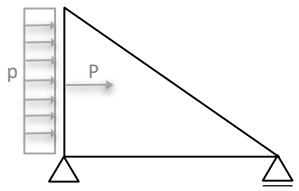
3. **چنانچه در سازه مسئله ۲ فنر عمودی Ks قرار گیرد، سازه را به روش اجزا محدود حل کنید. آیا تعادل برقرار است؟**



4. **همانند مسئله سوم، فنر به صورت زاویه دار قرار می گیرد.**

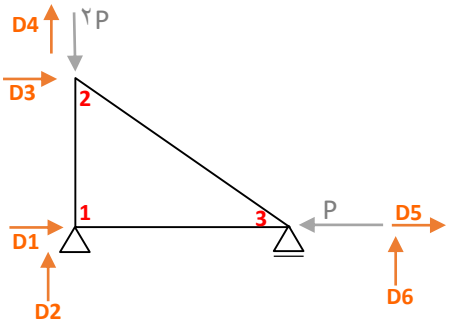


5. **چنانچه برای عضو ۲-۱ دو حالت بارگذاری زیر به صورت میان عضوی وارد شود، روش تحلیل اجزا محدود را به کار ببرید.**



حل:

.1



EL : 1-3 , θ=0 , S=0 , C=1 , AE , 4L

$$\{D_{El}\} = [R]\{D_{EG}\}$$

$$[R] = \begin{bmatrix} C & S & 0 & 0 \\ -S & C & 0 & 0 \\ 0 & 0 & C & S \\ 0 & 0 & -S & C \end{bmatrix} \rightarrow [R] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\{D_{EL}\} = [R]\{D_{EG}\} \rightarrow \{D_{EL}\} = [R]\{\overset{D1}{0}\,,\overset{D2}{0}\,,\overset{D5}{-4.00PL/AE}\,,\overset{D6}{0}\}^T$$

$$\{D_{EL}\} = \{0\,,0\,,-4PL/AE\,,0\}^T$$

$$\{P_{EL}\} = [S_{EL}]\{D_{EL}\} \quad , \quad [S_{EL}] = \frac{AE}{L}\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\{P_{EL}\} = \frac{AE}{4L}\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}\begin{Bmatrix} 0 \\ 0 \\ -4PL/AE \\ 0 \end{Bmatrix} = \begin{Bmatrix} P \\ 0 \\ -P \\ 0 \end{Bmatrix}$$

EL : 2-3 , θ=143.1 , S=0.6 , C=-0.8 , AE , 5L

$$\{D_{El}\} = [R]\{D_{EG}\}$$

$$[R] = \begin{bmatrix} C & S & 0 & 0 \\ -S & C & 0 & 0 \\ 0 & 0 & C & S \\ 0 & 0 & -S & C \end{bmatrix} \rightarrow [R] = \begin{bmatrix} -0.8 & 0.6 & 0 & 0 \\ -0.6 & -0.8 & 0 & 0 \\ 0 & 0 & -0.8 & 0.6 \\ 0 & 0 & -0.6 & -0.8 \end{bmatrix}$$

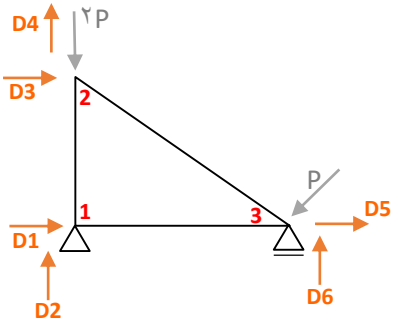
$$\{D_{EL}\} = [R]\{D_{EG}\} \quad \rightarrow \{D_{EL}\} = [R]\{\overset{D3}{-8.504PL/AE}\ , \overset{D4}{-6.006PL/AE}\ , \overset{D5}{-4.00PL/AE}\ , \overset{D6}{0}\}^T$$

$$\{D_{EL}\} = \{3.2PL/AE, 10PL/AE, 3.2PL/AE, 2.4PL/AE\}^T$$

$$\{P_{EL}\} = [S_{EL}]\{D_{EL}\} \quad , \quad [S_{EL}] = \frac{AE}{L}\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\{P_{EL}\} = \frac{AE}{5L}\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}\begin{Bmatrix} 3.2PL/AE \\ 10PL/AE \\ 3.2PL/AE \\ 2.4PL/AE \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{Bmatrix}$$

.2



$$[S]_{Structure} = \frac{AE}{L}\begin{array}{c} \begin{array}{ccc} D3 & D4 & D5 \end{array} \\ \begin{bmatrix} 0.128 & -0.096 & -0.128 \\ -0.096 & 0.405 & 0.096 \\ -0.128 & 0.096 & 0.378 \end{bmatrix} \end{array}$$

$$\{D_{EG}\} = \begin{Bmatrix} D_3 \\ D_4 \\ D_5 \end{Bmatrix} = [S]^{-1}\begin{Bmatrix} P_3 \\ P_4 \\ P_5 \end{Bmatrix} = [S]^{-1}\begin{Bmatrix} 0 \\ -2P \\ -\frac{\sqrt{2}}{2}P \end{Bmatrix} = \begin{Bmatrix} -7.333\frac{PL}{AE} \\ -6.006\frac{PL}{AE} \\ -2.828\frac{PL}{AE} \end{Bmatrix}$$

EL : 1-2 , θ=π/2 , S=1 , C=0 , AE , 3L

$$\{D_{El}\} = [R]\{D_{EG}\}$$

$$[R] = \begin{bmatrix} C & S & 0 & 0 \\ -S & C & 0 & 0 \\ 0 & 0 & C & S \\ 0 & 0 & -S & C \end{bmatrix} \rightarrow [R] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{bmatrix}$$

$$\{D_{EL}\} = [R]\{D_{EG}\} \rightarrow \{D_{EL}\} = [R]\{\overset{D1}{0}, \overset{D2}{0}, \overset{D3}{-7.333\frac{PL}{AE}}, \overset{D4}{-6.006\frac{PL}{AE}}\}^T$$

$$\{D_{EL}\} = \{0, 0, -6.006PL/AE, 7.333PL/AE\}^T$$

$$\{P_{EL}\} = [S_{EL}]\{D_{EL}\} \quad , \quad [S_{EL}] = \frac{AE}{L}\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\{P_{EL}\} = \frac{AE}{3L}\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}\begin{Bmatrix} 0 \\ 0 \\ -6.006\frac{PL}{AE} \\ 7.333\frac{PL}{AE} \end{Bmatrix} = \begin{Bmatrix} 2.002P \\ 0 \\ -2.002P \\ 0 \end{Bmatrix}$$

EL : 1-3 , θ=0 , S=0 , C=1 , AE , 4L

$$\{D_{El}\} = [R]\{D_{EG}\}$$

$$[R] = \begin{bmatrix} C & S & 0 & 0 \\ -S & C & 0 & 0 \\ 0 & 0 & C & S \\ 0 & 0 & -S & C \end{bmatrix} \rightarrow [R] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\{D_{EL}\} = [R]\{D_{EG}\} \rightarrow \{D_{EL}\} = [R]\{\overset{D1}{0}, \overset{D2}{0}, \overset{D5}{-2.828\frac{PL}{AE}}, \overset{D6}{0}\}^T$$

$$\{D_{EL}\} = \{0, 0, - - 2.828\frac{PL}{AE}, 0\}^T$$

$$\{P_{EL}\} = [S_{EL}]\{D_{EL}\} \quad , \quad [S_{EL}] = \frac{AE}{4L}\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\{P_{EL}\} = \frac{AE}{4L}\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}\begin{Bmatrix} 0 \\ 0 \\ -2.828\frac{PL}{AE} \\ 0 \end{Bmatrix} = \begin{Bmatrix} 0.707P \\ 0 \\ -0.707P \\ 0 \end{Bmatrix}$$

EL : 2-3 , θ=143.1 , S=0.6 , C=-0.8 , AE , 5L

$$\{D_{El}\} = [R]\{D_{EG}\}$$

$$[R] = \begin{bmatrix} C & S & 0 & 0 \\ -S & C & 0 & 0 \\ 0 & 0 & C & S \\ 0 & 0 & -S & C \end{bmatrix} \rightarrow [R] = \begin{bmatrix} -0.8 & 0.6 & 0 & 0 \\ -0.6 & -0.8 & 0 & 0 \\ 0 & 0 & -0.8 & 0.6 \\ 0 & 0 & -0.6 & -0.8 \end{bmatrix}$$

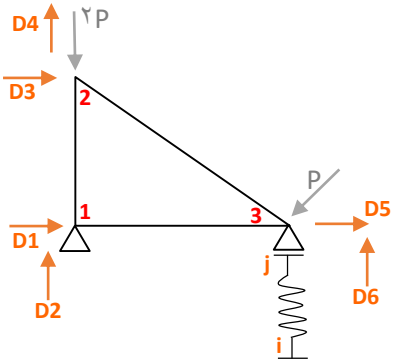
$$\{D_{EL}\} = [R]\{D_{EG}\} \rightarrow \{D_{EL}\} = [R]\{\overset{D3}{-7.333\frac{PL}{AE}}, \overset{D4}{-6.006\frac{PL}{AE}}, \overset{D5}{-2.828\frac{PL}{AE}}, \overset{D6}{0}\}^T$$

$$\{D_{EL}\} = \{2.263PL/AE, 9.205PL/AE, 2.265PL/AE, 1.697PL/AE\}^T$$

$$\{P_{EL}\} = [S_{EL}]\{D_{EL}\} \quad , \quad [S_{EL}] = \frac{AE}{L}\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\{P_{EL}\} = \frac{AE}{5L}\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}\begin{Bmatrix} 2.265PL/AE \\ 9.205PL/AE \\ 2.265PL/AE \\ 1.697PL/AE \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{Bmatrix}$$

.3



تعیین درجات آزادی

Node 1 : D1, D2 Support D.o.fs
Node 2 : D3, D4 Moveable D.o.fs
Node 3 : D5, D6 Moveable D.o.fs

تعیین موقعیت گره ها المان به المان

EL: 1-2 , θ=π/2 , S=1 , C=0 , AE , 3L

$$[S]_{1-2\ EG} = \frac{AE}{3L}\begin{array}{c} \begin{array}{cccc} D1 & D2 & D3 & D4 \end{array} \\ \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix} \end{array}$$

EL: 1-3 , θ=0 , S=0 , C=1 , AE , 4L

$$[S]_{1-3\ EG} = \frac{AE}{4L}\begin{array}{c} \begin{array}{cccc} D1 & D2 & D5 & D6 \end{array} \\ \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

EL: 2-3 , θ=143.1 , S=0.6 , C=-0.8 , AE , 5L

$$[S]_{2-3\ EG} = \frac{AE}{5L}\begin{array}{c} \begin{array}{cccc} D3 & D4 & D5 & D6 \end{array} \\ \begin{bmatrix} 0.64 & -0.48 & -0.64 & 0.48 \\ -0.48 & 0.36 & 0.48 & -0.36 \\ -0.64 & 0.48 & 0.64 & -0.48 \\ 0.48 & -0.36 & -0.48 & 0.36 \end{bmatrix} \end{array}$$

Spring , K= 5AE/L

$$[S]_{Spring\ EG} = \frac{5AE}{l}\begin{array}{c} \begin{array}{cccc} & & i & j \end{array} \\ \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix} \end{array}$$

سوار نمودن ماتریس سختی کل سازه

$$[S]_T = \frac{AE}{L}\begin{array}{c} \begin{array}{cccccc} D1 & D2 & D3 & D4 & D5 & D6 \end{array} \\ \begin{bmatrix} 0.25 & 0 & 0 & 0 & -0.25 & 0 \\ 0 & 0.333 & 0 & -0.333 & 0 & 0 \\ 0 & 0 & 0.128 & -0.096 & -0.128 & 0.096 \\ 0 & -0.333 & -0.096 & 0.405 & 0.096 & -0.072 \\ -0.25 & 0 & -0.128 & 0.096 & 0.378 & -0.096 \\ 0 & 0 & 0.096 & -0.072 & -0.096 & 5.072 \end{bmatrix} \end{array}$$

حذف درجات آزادی متناظر با درجات آزادی تکیه گاهی

$$[S]_{Structure} = \frac{AE}{L}\begin{bmatrix} 0.128 & -0.096 & -0.128 & 0.096 \\ -0.096 & 0.405 & 0.096 & -0.072 \\ -0.128 & 0.096 & 0.378 & -0.096 \\ 0.096 & -0.072 & -0.096 & 5.072 \end{bmatrix} \quad D_1 , D_2 = 0$$

تعیین بردار بارگذاری متناظر با درجات آزادی حرکت دار با توجه به رابطه $[S]\{D\} = \{P\}$

$$\{P\} = \begin{Bmatrix} 0 \\ -2P \\ -\frac{\sqrt{2}}{2}P \\ -\frac{\sqrt{2}}{2}P \end{Bmatrix}$$

محاسبه بردار تغییر مکان ها در سیستم Global

$$\{D_{EG}\} = \begin{Bmatrix} D_3 \\ D_4 \\ D_5 \\ D_6 \end{Bmatrix} = [S]^{-1}\begin{Bmatrix} P_3 \\ P_4 \\ P_5 \\ P_6 \end{Bmatrix} = [S]^{-1}\begin{Bmatrix} 0 \\ -2P \\ -\frac{\sqrt{2}}{2}P \\ -\frac{\sqrt{2}}{2}P \end{Bmatrix} = \begin{Bmatrix} -0.722PL/AE \\ -6.006PL/AE \\ 3.676PL/AE \\ -0.141PL/AE \end{Bmatrix}\begin{matrix} D3 \\ D4 \\ D5 \\ D6 \end{matrix}$$

محاسبه بردارهای تغییر مکان در سیستم Local و تعیین نیروهای داخلی هر یک از المان ها

EL: 1-2 , θ=π/2 , S=1 , C=0 , AE , 3L

$$\{D_{EL}\} = [R]\{D_{EG}\} \rightarrow \{D_{EL}\} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{bmatrix}\begin{Bmatrix} 0 \\ 0 \\ -0.722\frac{PL}{AE} \\ -6.006\frac{PL}{AE} \end{Bmatrix} \rightarrow \{D_{EL}\} = \begin{Bmatrix} 0 \\ 0 \\ -6.006\frac{PL}{AE} \\ 0.722\frac{PL}{AE} \end{Bmatrix}$$

$$\{P_{EL}\} = [S_{EL}]\{D_{EL}\} \rightarrow \frac{AE}{3L}\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}\begin{Bmatrix} 0 \\ 0 \\ -6.006\frac{PL}{AE} \\ 0.722\frac{PL}{AE} \end{Bmatrix} = \begin{Bmatrix} 2.002P \\ 0 \\ -2.002P \\ 0 \end{Bmatrix}$$

EL: 1-3 , θ=0 , S=0 , C=1 , AE , 4L

$$\{D_{EL}\} = [R]\{D_{EG}\} \rightarrow \{D_{EL}\} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} 0 \\ 0 \\ 3.676\dfrac{PL}{AE} \\ -0.141\dfrac{PL}{AE} \end{Bmatrix} \rightarrow \{D_{EL}\} = \begin{Bmatrix} 0 \\ 0 \\ 3.676\dfrac{PL}{AE} \\ -0.141\dfrac{PL}{AE} \end{Bmatrix}$$

$$\{P_{EL}\} = [S_{EL}]\{D_{EL}\} \rightarrow \frac{AE}{4L} \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{Bmatrix} 0 \\ 0 \\ 3.676\dfrac{PL}{AE} \\ -0.141\dfrac{PL}{AE} \end{Bmatrix} = \begin{Bmatrix} -0.92P \\ 0 \\ 0.92P \\ 0 \end{Bmatrix}$$

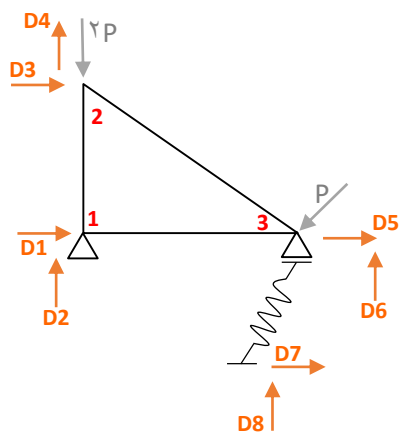
EL: 2-3 , θ=143.1 , S=0.6 , C=-0.8 , AE , 5L

$$\{D_{EL}\} = [R]\{D_{EG}\} \rightarrow \{D_{EL}\} = \begin{bmatrix} -0.8 & 0.6 & 0 & 0 \\ -0.6 & -0.8 & 0 & 0 \\ 0 & 0 & -0.8 & 0.6 \\ 0 & 0 & -0.6 & -0.8 \end{bmatrix} \begin{Bmatrix} -0.722\dfrac{PL}{AE} \\ -6.006\dfrac{PL}{AE} \\ 3.676\dfrac{PL}{AE} \\ -0.141\dfrac{PL}{AE} \end{Bmatrix} = \begin{Bmatrix} -3.026\dfrac{PL}{AE} \\ 5.238\dfrac{PL}{AE} \\ -3.026\dfrac{PL}{AE} \\ -2.093\dfrac{PL}{AE} \end{Bmatrix}$$

$$\{P_{EL}\} = [S_{EL}]\{D_{EL}\} \rightarrow \frac{AE}{5L} \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{Bmatrix} -3.026\dfrac{PL}{AE} \\ 5.238\dfrac{PL}{AE} \\ -3.026\dfrac{PL}{AE} \\ -2.093\dfrac{PL}{AE} \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{Bmatrix}$$

در تمامی المان ها تعادل برقرار است.

.4



تعیین درجات آزادی

Node 1 : D1, D2 Support D.o.fs
Node 2 : D3, D4 Moveable D.o.fs
Node 3 : D5, D6 Moveable D.o.fs

تعیین موقعیت گره ها المان به المان

EL: 1-2 , θ=π/2 , S=1 , C=0 , AE , 3L

$$[S]_{1-2\ EG} = \frac{AE}{3L}\begin{array}{c} \begin{array}{cccc} D1 & D2 & D3 & D4 \end{array} \\ \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix} \end{array}$$

EL: 1-3 , θ=0 , S=0 , C=1 , AE , 4L

$$[S]_{1-3\ EG} = \frac{AE}{4L}\begin{array}{c} \begin{array}{cccc} D1 & D2 & D5 & D6 \end{array} \\ \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

EL: 2-3 , θ=143.1 , S=0.6 , C=-0.8 , AE , 5L

$$[S]_{2-3\ EG} = \frac{AE}{5L}\begin{array}{c} \begin{array}{cccc} D3 & D4 & D5 & D6 \end{array} \\ \begin{bmatrix} 0.64 & -0.48 & -0.64 & 0.48 \\ -0.48 & 0.36 & 0.48 & -0.36 \\ -0.64 & 0.48 & 0.64 & -0.48 \\ 0.48 & -0.36 & -0.48 & 0.36 \end{bmatrix} \end{array}$$

Spring , cosθ=C , sinθ=S , K= 5AE/L

$$[S]_{Spring\ EG} = \frac{5AE}{l}\begin{array}{c} \begin{array}{cccc} D7 & D8 & D5 & D6 \end{array} \\ \begin{bmatrix} C^2 & CS & -C^2 & -CS \\ CS & S^2 & -CS & -S^2 \\ -C^2 & -CS & C^2 & CS \\ -CS & -S^2 & CS & S^2 \end{bmatrix} \end{array}$$

سوار نمودن ماتریس سختی کل سازه

$$
[S]_T = \frac{AE}{L}\begin{array}{c} \begin{array}{cccccc} D1 & D2 & D3 & D4 & D5 & D6 \end{array} \\ \begin{bmatrix} 0.25 & 0 & 0 & 0 & -0.25 & 0 \\ 0 & 0.333 & 0 & -0.333 & 0 & 0 \\ 0 & 0 & 0.128 & -0.096 & -0.128 & 0.096 \\ 0 & -0.333 & -0.096 & 0.405 & 0.096 & -0.072 \\ -0.25 & 0 & -0.128 & 0.096 & 0.378+5C^2 & -0.096+5CS \\ 0 & 0 & 0.096 & -0.072 & -0.096+5CS & 0.072+5S^2 \end{bmatrix} \end{array}
$$

حذف درجات آزادی متناظر با درجات آزادی تکیه گاهی

$$
[S]_{Structure} = \frac{AE}{L}\begin{bmatrix} 0.128 & -0.096 & -0.128 & 0.096 \\ -0.096 & 0.405 & 0.096 & -0.072 \\ -0.128 & 0.096 & 0.378+5C^2 & -0.096+5CS \\ 0.096 & -0.072 & -0.096+5CS & 0.072+5S^2 \end{bmatrix} \quad D_1\,, D_2 = 0
$$

تعیین بردار بارگذاری متناظر با درجات آزادی حرکت دار با توجه به رابطه $[S]\{D\} = \{P\}$

$$
\{P\} = \begin{Bmatrix} 0 \\ -2P \\ -\frac{\sqrt{2}}{2}P \\ -\frac{\sqrt{2}}{2}P \end{Bmatrix}
$$

محاسبه بردار تغییر مکان ها در سیستم Global

$$
\{D_{EG}\} = \begin{Bmatrix} D_3 \\ D_4 \\ D_5 \\ D_6 \end{Bmatrix} = [S]^{-1}\begin{Bmatrix} P_3 \\ P_4 \\ P_5 \\ P_6 \end{Bmatrix} = [S]^{-1}\begin{Bmatrix} 0 \\ -2P \\ -\frac{\sqrt{2}}{2}P \\ -\frac{\sqrt{2}}{2}P \end{Bmatrix} = \frac{PL}{AE}\begin{Bmatrix} \frac{\sqrt{2}(60C^2+80CS+3)}{40S^2} - \frac{\sqrt{2}(3C+4S)}{2S} - \frac{500}{111} \\ -6.006 \\ \frac{2\sqrt{2}C}{S} - 2\sqrt{2} \\ \frac{2\sqrt{2}C}{S} - \frac{\sqrt{2}(20C^2+1)}{10S^2} \end{Bmatrix} \begin{array}{l} D3 \\ D4 \\ D5 \\ D6 \end{array}
$$

محاسبه بردارهای تغییر مکان در سیستم Local و تعیین نیروهای داخلی هر یک از المان ها

EL: 1-2 , θ=π/2 , S=1 , C=0 , AE , 3L

$$
\{D_{EL}\} = [R]\{D_{EG}\} \rightarrow \{D_{EL}\} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{bmatrix}\begin{Bmatrix} 0 \\ 0 \\ D3 \\ -6.006\frac{PL}{AE} \end{Bmatrix} \rightarrow \{D_{EL}\} = \begin{Bmatrix} 0 \\ 0 \\ -6.006\frac{PL}{AE} \\ -\left(\frac{\sqrt{2}(60C^2+80CS+3)}{40S^2} - \frac{\sqrt{2}(3C+4S)}{2S} - \frac{500}{111}\right)\frac{PL}{AE} \end{Bmatrix}
$$

$$
\{P_{EL}\} = [S_{EL}]\{D_{EL}\} \rightarrow \frac{AE}{3L}\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}\begin{Bmatrix} 0 \\ 0 \\ -6.006\frac{PL}{AE} \\ D3 \end{Bmatrix} = \begin{Bmatrix} 2.002P \\ 0 \\ -2.002P \\ 0 \end{Bmatrix}
$$

EL: 1-3 , θ=0 , S=0 , C=1 , AE , 4L

$$\{D_{EL}\} = [R]\{D_{EG}\} \rightarrow \{D_{EL}\} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} 0 \\ 0 \\ \left(\frac{2\sqrt{2}C}{S} - 2\sqrt{2}\right)\frac{PL}{AE} \\ \left(\frac{2\sqrt{2}C}{S} - \frac{\sqrt{2}(20C^2 + 1)}{10S^2}\right)\frac{PL}{AE} \end{Bmatrix} \rightarrow \{D_{EL}\} = \begin{Bmatrix} 0 \\ 0 \\ \left(\frac{2\sqrt{2}C}{S} - 2\sqrt{2}\right)\frac{PL}{AE} \\ \left(\frac{2\sqrt{2}C}{S} - \frac{\sqrt{2}(20C^2 + 1)}{10S^2}\right)\frac{PL}{AE} \end{Bmatrix}$$

$$\{P_{EL}\} = [S_{EL}]\{D_{EL}\} \rightarrow \frac{AE}{4L}\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{Bmatrix} 0 \\ 0 \\ \left(\frac{2\sqrt{2}C}{S} - 2\sqrt{2}\right)\frac{PL}{AE} \\ \left(\frac{2\sqrt{2}C}{S} - \frac{\sqrt{2}(20C^2 + 1)}{10S^2}\right)\frac{PL}{AE} \end{Bmatrix} = \begin{Bmatrix} -\left(\frac{\sqrt{2}C}{2S} - \frac{\sqrt{2}}{2}\right)P \\ 0 \\ \left(\frac{\sqrt{2}C}{2S} - \frac{\sqrt{2}}{2}\right)P \\ 0 \end{Bmatrix}$$

EL: 2-3 , θ=143.1 , S=0.6 , C=-0.8 , AE , 5L

$$\{D_{EL}\} = [R]\{D_{EG}\} \rightarrow \{D_{EL}\} = \begin{bmatrix} -0.8 & 0.6 & 0 & 0 \\ -0.6 & -0.8 & 0 & 0 \\ 0 & 0 & -0.8 & 0.6 \\ 0 & 0 & -0.6 & -0.8 \end{bmatrix}\frac{PL}{AE} \begin{Bmatrix} \frac{\sqrt{2}(60C^2 + 80CS + 3)}{40S^2} - \frac{\sqrt{2}(3C + 4S)}{2S} - \frac{500}{111} \\ -6.006 \\ \frac{2\sqrt{2}C}{S} - 2\sqrt{2} \\ \frac{2\sqrt{2}C}{S} - \frac{\sqrt{2}(20C^2 + 1)}{10S^2} \end{Bmatrix}$$

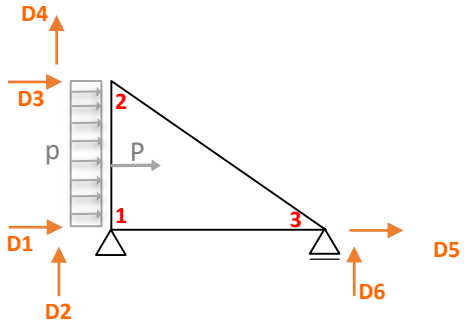
$$= \frac{PL}{AE}\begin{Bmatrix} -0.3\frac{(60C^2 + 80CS + 3)}{S^2} + 0.56\frac{(3C + 4S)}{S} \\ -0.02\frac{(60C^2 + 80CS + 3)}{S^2} + 0.42\frac{(3C + 4S)}{S} + 7.058 \\ -2.26\left(\frac{C}{S} - 1\right) + \left(\frac{1.7C}{S} - 0.08\frac{(20C^2 + 1)}{S^2}\right) \\ -1.7\left(\frac{C}{S} - 1\right) - \left(2.26\frac{C}{S} - 0.11\frac{(20C^2 + 1)}{S^2}\right) \end{Bmatrix}$$

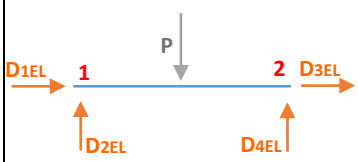
$$\{P_{EL}\} = [S_{EL}]\{D_{EL}\} \rightarrow \frac{AE}{5L}\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}\frac{PL}{AE} \begin{Bmatrix} -0.3\frac{(60C^2 + 80CS + 3)}{S^2} + 0.56\frac{(3C + 4S)}{S} \\ -0.02\frac{(60C^2 + 80CS + 3)}{S^2} + 0.42\frac{(3C + 4S)}{S} + 7.058 \\ -2.26\left(\frac{C}{S} - 1\right) + \left(\frac{1.7C}{S} - 0.08\frac{(20C^2 + 1)}{S^2}\right) \\ -1.7\left(\frac{C}{S} - 1\right) - \left(2.26\frac{C}{S} - 0.11\frac{(20C^2 + 1)}{S^2}\right) \end{Bmatrix}$$

$$= \begin{Bmatrix} \left(21.76\frac{C}{S} - 16.4\frac{C^2}{S^2} - \frac{0.82}{S^2}\right)P \\ 0 \\ -\left(21.76\frac{C}{S} - 16.4\frac{C^2}{S^2} - \frac{0.82}{S^2}\right)P \\ 0 \end{Bmatrix}$$

در تمامی المان ها تعادل برقرار است.

.5





تعیین بار معادل گره ای در مختصات Global

تابع شکل نقاط:

$$N_1 = 1 - \frac{x}{l}\ ,\ \ N_2 = \frac{x}{l}$$

بار متمرکز:

$$\{P_{EL}\} = \begin{Bmatrix} P_{2\,EL} = -P * (1 - \frac{3\,l/2}{3l}) = -\frac{P}{2} \\ P_{4\,EL} = -P * (\frac{3\,l/2}{3l}) = -\frac{P}{2} \end{Bmatrix}$$

$$\{P_{EG}\} = [R]^T\{P_{EL}\} \rightarrow \{P_{EG}\} = \begin{Bmatrix} P_{1\,EG} = 0 \\ P_{2\,EG} = P/2 \\ P_{3\,EG} = 0 \\ P_{4\,EG} = P/2 \end{Bmatrix}$$

بار گسترده:

$$\{P_{EL}\} = \begin{Bmatrix} P_{2\,EL} = \int_0^{3L} -p * (1 - \frac{x}{3l}) = -\frac{3P}{2} \\ P_{4\,EL} = \int_0^{3L} -p * (\frac{x}{3l}) = -\frac{3P}{2} \end{Bmatrix}$$

$$\{P_{EG}\} = [R]^T\{P_{EL}\} \rightarrow \{P_{EG}\} = \begin{Bmatrix} P_{1\,EG} = 0 \\ P_{2\,EG} = 3P/2 \\ P_{3\,EG} = 0 \\ P_{4\,EG} = 3P/2 \end{Bmatrix}$$

در نتیجه مقدار 2P بار به گره ۱ و ۲ در جهت مثبت x وارد می شود. به علت مقید بودن گره اول، جابجایی متناظر با بار وارد شده در این گره وجود ندارد و فقط جابجایی بار معادل گره ای Node 2 بررسی می شود.

تعیین درجات آزادی

Node 1 : D1, D2 Support D.o.fs
Node 2 : D3, D4 Moveable D.o.fs
Node 3 : D5, D6 : Moveable, Support D.o.fs

تعیین موقعیت گره ها المان به المان

EL: 1-2 , θ=π/2 , S=1 , C=0 , AE , 3L

$$[S]_{1-2\ EG} = \frac{AE}{3L}\begin{array}{c} \begin{array}{cccc} D1 & D2 & D3 & D4 \end{array} \\ \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix} \end{array}$$

EL: 1-3 , θ=0 , S=0 , C=1 , AE , 4L

$$[S]_{1-3\ EG} = \frac{AE}{4L}\begin{array}{c} \begin{array}{cccc} D1 & D2 & D5 & D6 \end{array} \\ \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

EL: 2-3 , θ=143.1 , S=0.6 , C=-0.8 , AE , 5L

$$[S]_{2-3\ EG} = \frac{AE}{5L}\begin{array}{c} \begin{array}{cccc} D3 & D4 & D5 & D6 \end{array} \\ \begin{bmatrix} 0.64 & -0.48 & -0.64 & 0.48 \\ -0.48 & 0.36 & 0.48 & -0.36 \\ -0.64 & 0.48 & 0.64 & -0.48 \\ 0.48 & -0.36 & -0.48 & 0.36 \end{bmatrix} \end{array}$$

سوار نمودن ماتریس سختی کل سازه

$$[S]_T = \frac{AE}{L}\begin{array}{c} \begin{array}{cccccc} D1 & D2 & D3 & D4 & D5 & D6 \end{array} \\ \begin{bmatrix} 0.25 & 0 & 0 & 0 & -0.25 & 0 \\ 0 & 0.333 & 0 & -0.333 & 0 & 0 \\ 0 & 0 & 0.128 & -0.096 & -0.128 & 0.096 \\ 0 & -0.333 & -0.096 & 0.405 & 0.096 & -0.072 \\ -0.25 & 0 & -0.128 & 0.096 & 0.378 & -0.096 \\ 0 & 0 & 0.096 & -0.072 & -0.096 & 0.072 \end{bmatrix} \end{array}$$

حذف درجات آزادی متناظر با درجات آزادی تکیه گاهی

$$[S]_{Structure} = \frac{AE}{L}\begin{bmatrix} 0.128 & -0.096 & -0.128 \\ -0.096 & 0.405 & 0.096 \\ -0.128 & 0.096 & 0.378 \\ 0.096 & -0.072 & -0.096 \end{bmatrix} \quad D_1, D_2, D_6 = 0$$

تعیین بردار بارگذاری متناظر با درجات آزادی حرکت دار با توجه به رابطه $[S]\{D\} = \{P\}$

$$\{P\} = \begin{Bmatrix} 2P \\ 0 \\ 0 \end{Bmatrix}\begin{array}{c} D3 \\ D4 \\ D5 \end{array}$$

محاسبه بردار تغییر مکان ها در سیستم Global

$$\{D_{EG}\} = \begin{Bmatrix} D_3 \\ D_4 \\ D_5 \end{Bmatrix} = [S]^{-1}\begin{Bmatrix} P_3 \\ P_4 \\ P_5 \end{Bmatrix} = [S]^{-1}\begin{Bmatrix} 2P \\ 0 \\ 0 \end{Bmatrix} = \frac{PL}{AE}\begin{Bmatrix} 27.0034 \\ 4.5045 \\ 8 \end{Bmatrix}\begin{matrix} D3 \\ D4 \\ D5 \end{matrix}$$

محاسبه بردارهای تغییر مکان در سیستم Local و تعیین نیروهای داخلی هر یک از المان ها

EL : 1-2 , θ=π/2 , S=1 , C=0 , AE , 3L

$$\{D_{El}\} = [R]\{D_{EG}\}$$

$$[R] = \begin{bmatrix} C & S & 0 & 0 \\ -S & C & 0 & 0 \\ 0 & 0 & C & S \\ 0 & 0 & -S & C \end{bmatrix} \rightarrow [R] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{bmatrix}$$

$$\{D_{EL}\} = [R]\{D_{EG}\} \rightarrow \{D_{EL}\} = [R]\begin{Bmatrix} 0 \\ 0 \\ 27.0034\dfrac{PL}{AE} \\ 4.5045\dfrac{PL}{AE} \end{Bmatrix} \rightarrow \{D_{EL}\} = \begin{Bmatrix} 0 \\ 0 \\ 4.5045\dfrac{PL}{AE} \\ -27.0034\dfrac{PL}{AE} \end{Bmatrix}\begin{matrix} D1 \\ D2 \\ D3 \\ D4 \end{matrix}$$

$$\{P_{EL}\} = [S_{EL}]\{D_{EL}\} \rightarrow \{P_{EL}\} = \frac{AE}{3L}\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}\begin{Bmatrix} 0 \\ 0 \\ 4.5045\dfrac{PL}{AE} \\ -27.0034\dfrac{PL}{AE} \end{Bmatrix} = \begin{Bmatrix} -1.168P \\ 0 \\ 1.168P \\ 0 \end{Bmatrix}$$

EL : 1-3 , θ=0 , S=0 , C=1 , AE , 4L

$$\{D_{El}\} = [R]\{D_{EG}\}$$

$$[R] = \begin{bmatrix} C & S & 0 & 0 \\ -S & C & 0 & 0 \\ 0 & 0 & C & S \\ 0 & 0 & -S & C \end{bmatrix} \rightarrow [R] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\{D_{EL}\} = [R]\{D_{EG}\} \rightarrow \{D_{EL}\} = [R]\begin{Bmatrix} 0 \\ 0 \\ 8\dfrac{PL}{AE} \\ 0 \end{Bmatrix} \rightarrow \{D_{EL}\} = \begin{Bmatrix} 0 \\ 0 \\ 8\,PL/AE \\ 0 \end{Bmatrix}\begin{matrix} D1 \\ D2 \\ D5 \\ D6 \end{matrix}$$

$$\{P_{EL}\} = [S_{EL}]\{D_{EL}\} \rightarrow \{P_{EL}\} = \frac{AE}{4L}\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}\begin{Bmatrix} 0 \\ 0 \\ 8\dfrac{PL}{AE} \\ 0 \end{Bmatrix} = \begin{Bmatrix} -2P \\ 0 \\ 2P \\ 0 \end{Bmatrix}$$

EL : 2-3 , θ=143.1 , S=0.6 , C=-0.8 , AE , 5L

$$\{D_{El}\} = [R]\{D_{EG}\}$$

$$[R] = \begin{bmatrix} C & S & 0 & 0 \\ -S & C & 0 & 0 \\ 0 & 0 & C & S \\ 0 & 0 & -S & C \end{bmatrix} \rightarrow [R] = \begin{bmatrix} -0.8 & 0.6 & 0 & 0 \\ -0.6 & -0.8 & 0 & 0 \\ 0 & 0 & -0.8 & 0.6 \\ 0 & 0 & -0.6 & -0.8 \end{bmatrix}$$

$$\{D_{EL}\} = [R]\{D_{EG}\} \rightarrow \{D_{EL}\} = [R]\begin{Bmatrix} 27.0034\ PL/AE \\ 4.5045\ PL/AE \\ 8\ \ PL/AE \\ 0 \end{Bmatrix}\begin{matrix} \boldsymbol{D3} \\ \boldsymbol{D4} \\ \boldsymbol{D5} \\ \boldsymbol{D6} \end{matrix} \rightarrow \{D_{EL}\} = \begin{Bmatrix} -18.9\ PL/AE \\ -19.8\ PL/AE \\ -6.4\ \ PL/AE \\ -4.8\ \ PL/AE \end{Bmatrix}$$

$$\{P_{EL}\} = [S_{EL}]\{D_{EL}\} \rightarrow \{P_{EL}\} = \frac{AE}{5L}\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}\begin{Bmatrix} -18.9\ PL/AE \\ -19.8\ PL/AE \\ -6.4\ \ PL/AE \\ -4.8\ \ PL/AE \end{Bmatrix} = \begin{Bmatrix} -12.5\ P \\ 0 \\ 12.5\ P \\ 0 \end{Bmatrix}$$